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System and method of detecting a dragging brake in an elevator application

Abstract

A system and method for detecting a brake dragging during normal operation of a motor monitors torque at zero speed and at constant speed. For an inertial system acted upon by gravity, a value of torque required to maintain zero speed is approximately the same as a value of torque required to maintain operation at a constant speed. In an exemplary, elevator system, a motor drive determines the torque required to maintain zero speed operation of an elevator cab after a holding brake opens and before it begins controlling the motor to rotate. The motor drive again determines the torque required to maintain a constant speed and compares this value to the value of torque required to maintain zero speed. If the difference between the two values of torque is greater than a predefined threshold, then the motor drive determines that the brake is dragging and sets an error message.

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Background/Summary

BACKGROUND INFORMATION

- (1) The subject matter disclosed herein relates to a system and method for detecting a dragging brake in an elevator application and, more specifically, to a system and method for monitoring torque during operation of a motor controlling an elevator cab to determine proper operation of a holding brake in the elevator drive train.
- (2) Elevators are typically implemented as counterbalancing systems. Accordingly, cables, also referred to as ropes, are typically provided around a sheave (a grooved spindle or pulley), mounted to the top of an elevator cab in order to raise or lower the cab within an elevator shaft. A first end of the cables may be mounted to a first point at the top of the shaft and routed down and around a sheave mounted to the top of the cab. The cables may then be routed up over a sheave mounted to an electric motor with a drive shaft there between. The cables may then continue around one or more sheaves mounted to a counterweight, then back to a second point at the top of the shaft. Use of the counterweight provides counterbalancing with the cab which may permit the motor to lift a differential weight between the counterweight and the cab (as opposed to the entire weight of the cab). Various configurations of cables, sheaves, and cable routing may be utilized.
- (3) To ease strain on the motor and provide further safety in the system, a holding brake is typically used in conjunction with the motor. Operation of a holding brake is fundamentally different than a brake used to slow or stop a rotating motor, such as in a vehicular application. A motor drive controls operation of the motor to bring the motor to a stop, or nearly to a stop, prior to setting the brake. When the elevator cab is brought to a stop, the motor drive is commanded to maintain operation of the motor at zero speed as the brake is set. The motor drive may subsequently be disabled after the brake is engaged to hold the sheave, cables, cab and counterweight stationary. To resume motion, such as upon a call for the elevator to move to another floor, the motor drive is enabled and again commanded to hold the motor at zero speed. The brake is released and the motor drive may then be commanded to control the motor to move the cab as desired.
- (4) For safety, the holding brake is configured to normally engage the elevator drive train, applying a braking force when no power is supplied to the motor. It is contemplated that the holding brake may be connected at a shaft of the motor or on an external rotating member of the elevator drive train, such as the drive sheave. A spring applies a force against a brake member, such as brake pads, which, in turn, engage the rotating member of the elevator drive train. An actuator is energized to compress the spring. Power must be applied to the actuator to compress the spring and to release the holding brake, allowing rotation of the motor. In the event of a power failure or prior to energization of the system, the actuator is deenergized, causing the brake to set and stopping motion of the elevator cab.
- (5) As is known to those skilled in the art, a holding brake for an elevator commonly includes adjustments to optimize operation of the brake. The available adjustments may vary the force applied by the spring, the force applied by a solenoid to open the brake, a holding force applied by the solenoid after the brake is open, or a holding torque applied by the brake to the elevator drive train. During commissioning of an elevator, operation of the holding brake is adjusted to obtain desired operation. Over time or due to varying operating conditions, however, the brake may go out of adjustment, deviating from desired operation. This may cause the brake to not fully open (also referred to as a dragging brake) which, in turn, applies some friction force to the drive train during normal operation. The friction force may cause increased heating, excessive wear in brake surfaces, energy losses in the elevator system and, potentially, lead to the brake being unable to hold the

elevator cab when the motor is not providing a holding torque.

(6) Thus, it would be desirable to detect a brake dragging during normal operation of a motor driving the system in which the holding brake is installed.

BRIEF DESCRIPTION

(7) According to one embodiment of the invention, a system for detecting a dragging brake in an elevator drive includes an elevator brake, a motor, and a motor drive. The elevator drive may further include an elevator sheave and/or a gearbox operatively connected to the motor. The elevator brake includes at least one braking surface configured to engage a rotating member of the elevator drive to prevent rotation of the elevator drive and a spring configured to apply a force to the at least one braking surface, causing the at least one braking surface to engage the rotating member of the elevator drive. The elevator brake also includes an actuator selectively activated to apply a counter force to the spring. When the counter force is applied to the spring, the at least one braking surface disengages the rotating member of the elevator drive, allowing rotation of the elevator drive. The motor is operably connected to cause rotation of the elevator drive, and the motor drive is configured to control operation of the motor. The motor drive further includes a memory configured to store instructions and a processor configured to execute the instructions stored on the memory. The processor is configured to determine a first value of torque when the actuator of the elevator brake initially disengages the at least one braking surface from the rotating member of the elevator drive and prior to causing rotation of the elevator drive and to determine a second value of torque when the motor is rotating at a constant speed. The processor compares the first value of torque to the second value of torque to detect when the at least one braking surface of the elevator brake does not fully disengage the rotating member of the elevator drive during rotation of the elevator drive.

(8) According to another aspect of the invention, the motor drive is configured to generate a torque reference value corresponding to a desired level of torque supplied by the motor, where the first and second values of torque may be torque reference values. Optionally, the motor drive further comprises at least one current sensor configured to generate a current feedback signal corresponding to current output from the motor drive to the motor, and the first and second values of torque may be determined as a function of the current feedback signal.

(9) According to still another aspect of the invention, comparing the first value of torque to the second value of torque includes determining a difference between the first and second values of torque. Detecting when the at least one braking surface of the elevator brake does not fully disengage the rotating member of the elevator drive during rotation of the elevator drive may include comparing the difference to a predetermined threshold and detecting the dragging brake when the difference is greater than the predetermined threshold. Optionally, the motor drive may be further configured to initially determine an offset value between the first value of torque and the second value of torque. Detecting when the at least one braking surface of the elevator brake does not fully disengage the rotating member of the elevator drive during rotation of the elevator drive may include comparing the difference to the offset value and detecting the dragging brake when the difference is greater than the offset value.

(10) According to another embodiment of the invention, a system for detecting a dragging brake for an electric motor includes a motor and a motor drive. The motor is configured to control motion of a drive train for a load having an additional force acting on the load to cause motion of the load when not controlled by the drive train, and the motor drive is operatively connected to the motor. The motor drive is configured to determine a first value of torque when a holding brake in the drive train is initially released and prior to causing motion of the drive train with the motor. The motor drive is also configured to determine a second value of torque when the motor is rotating at a constant speed and to compare the first value of torque to the second value of torque to detect when the holding brake does not fully disengage the drive train during operation of the motor.

(11) According to yet another aspect of the invention, the motor drive further comprises an input

configured to receive a position feedback signal, and the position feedback signal is generated by a position feedback device operatively connected to the motor. The position feedback signal corresponds to an angular position of the motor, and the motor drive is configured to control operation of the motor to maintain the load at an initial position determined from the position feedback signal when the holding brake is released and until the first value of torque is determined. (12) According to yet another embodiment of the invention, a method for detecting a dragging brake for an electric motor is disclosed. A first value of torque, generated in a motor operatively connected to a motor drive, is determined when a holding brake in a drive train between the motor and a load controlled by the motor is initially released. The motor drive is configured to determine the first value of torque, and the load has an additional force acting on the load to cause motion of the load when not controlled by the drive train. A second value of torque is determined when the motor is rotating at a constant speed, and the first value of torque is compared to the second value of torque to detect the dragging brake.

(13) These and other advantages and features of the invention will become apparent to those skilled in the art from the detailed description and the accompanying drawings. It should be understood, however, that the detailed description and accompanying drawings, while indicating preferred embodiments of the present invention, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the present invention without departing from the spirit thereof, and the invention includes all such modifications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various exemplary embodiments of the subject matter disclosed herein are illustrated in the accompanying drawings in which like reference numerals represent like parts throughout, and in which:

(2) FIG. 1 is an exemplary elevator system including an elevator cab and a counterweight in which the system provides an electrical motor, a brake, an encoder and a controller in accordance with an embodiment of the invention;

(3) FIG. 2 is a block diagram representation of a motor drive operative to control the electrical motor of FIG. 1;

(4) FIG. 3 is a partial isometric view of an exemplary encoder which may be used in the system of FIG. 1;

(5) FIG. 4 is a partial cutaway view of an exemplary brake which may be used in the system of FIG. 1 which includes a toothed hub and a grooved rotor in which engagement of the toothed hub with the grooved rotor may provide a brake in the system;

(6) FIG. 5 is a perspective view of another exemplary brake which may be used in the system of FIG. 1 which includes brake pads configured to engage an exterior rotating surface;

(7) FIG. 6 is a flow diagram illustrating one embodiment of the present invention;

(8) FIG. 7A is a graphical representation of angular velocity of a motor during forward and reverse operation; and

(9) FIG. 7B is a graphical representation of torque produced during the forward and reverse operation shown in FIG. 7A both with and without a dragging brake.

(10) In describing the various embodiments of the invention which are illustrated in the drawings, specific terminology will be resorted to for the sake of clarity. However, it is not intended that the invention be limited to the specific terms so selected and it is understood that each specific term includes all technical equivalents which operate in a similar manner to accomplish a similar purpose. For example, the word “connected,” “attached,” or terms similar thereto are often used. They are not limited to direct connection but include connection through other elements where such

connection is recognized as being equivalent by those skilled in the art.

DETAILED DESCRIPTION

(11) The various features and advantageous details of the subject matter disclosed herein are explained more fully with reference to the non-limiting embodiments described in detail in the following description.

(12) The subject matter disclosed herein describes a system and method to detect a brake dragging during normal operation of a motor driving the system in which the holding brake is installed. A motor drive is configured to control operation of a motor for a drive train, which is, in turn, connected to a load having a force applied to the load in addition to forces applied by the drive train. The force may be, for example, gravitational or the result of another motor in the system. In particular, the present system and method is well adapted to detect a holding brake which is dragging in an elevator application. A typical elevator application includes a cab and a counter weight connected together via cables, also referred to as ropes. A holding brake engages a drive train to prevent undesired motion of the cab when the motor is not energized, and the motor is connected via the drive train to a sheave to drive the ropes and, in turn, drive the elevator cab to a desired floor.

(13) A single cycle of operation for an elevator begins with an elevator cab stopped at a floor of the building in which it operates. A call from another floor or a signal from within the elevator cab to move to another floor initiates a run. A controller enables the motor drive and issues a command to the motor drive to begin operating at zero speed. When the motor drive is able to provide a holding torque to the motor, the controller generates a command to the holding brake to open. An actuator on the holding brake applies a force counter to the spring force releasing the brake pad, or other braking surface, from the elevator drive train. Once the brake is released and prior to beginning motion of the elevator cab, the motor drive is supplying a current to the motor which provides the required holding torque in the motor to maintain zero speed operation without relying on the holding brake. The motor drive is configured to determine the amount of torque required to maintain zero speed operation before it begins causing the motor to rotate. As the motor begins to move, the motor is accelerated from zero speed to a constant running speed. During acceleration, additional current beyond that required to hold the load at zero speed is required to accelerate the motor and, in turn, accelerate the elevator cab up to speed. Once the motor has reached the maximum desired operating speed, the motor continues running at that constant speed.

(14) In an ideal, balanced elevator system, the amount of torque required to maintain constant speed operation during this constant speed operation would be nearly identical to the amount of torque required to hold the motor at zero speed without the holding brake set, and the motor would only be required to generate torque for acceleration or deceleration between constant speed and zero speed operation due to the inertia of the system. Although some inefficiencies in a non-ideal system, such as friction from the ropes or in a gearbox, may require some additional torque to maintain constant speed operation, the torque required to maintain operation at constant speed in a balanced system is still approximately the same as that required to maintain zero speed. The motor drive determines the torque required to maintain a constant speed and compares this value to the value of torque required to maintain zero speed. If the difference between the two values of torque is greater than a predefined threshold, this indicates there is an unexpected force, such as friction from a dragging brake, slowing the system that the motor must overcome. As a result the motor drive determines that the brake is dragging and sets an error message.

(15) Turning initially to FIG. 1, an exemplary elevator system **10** is illustrated in accordance with an embodiment of the invention. A shaft **12** includes a cab **14** configured to move up and down the shaft **12**. The cab **14** includes, for example, wheels configured to engage rails **16** extending vertically along each side of the shaft **12** to maintain horizontal alignment of the cab **14** within the shaft **12**. Cables **20** extending around one or more cab sheaves **18** mounted to the top of the cab **14** may be used to raise or lower the cab **14** within the shaft **12**. According to the illustrated

embodiment, a first end of the cables **20** is fixedly mounted to a first point at the top of the shaft **12** and routed down and around the cab sheave **18** mounted to the top of the cab **14**. The cables **20** are then routed up and over one or more drive sheaves **78** mounted to an electrical motor **70**. The cables **20** continue around one or more counterweight sheaves **32** mounted to a counterweight **30** and back to a second point at the top of the shaft **12**. It is contemplated that various other configurations of cables, sheaves, and cable routing may be utilized according to the application requirements without deviating from the scope of the invention.

(16) According to the illustrated embodiment, the motor **70** may be mounted in a machine room located above the elevator shaft **12**. The motor **70** may be connected via a gearbox to the drive sheave **78**, where the motor, gearbox, and drive sheave form the elevator drive train. Optionally, the motor **70** and drive sheave **78** may be formed as a single unit, where the drive sheave is, at least in part, the rotor of the motor, forming a direct-drive system. In some applications, and, particularly with a direct drive system, the motor **70** may be mounted in the elevator shaft **12**. A brake **60**, is operatively connected to the elevator drive train to prevent rotation. The brake **60** may be connected to an output shaft of the motor **70** or to the sheave **78**. An encoder **80** is operatively connected to the motor **70** to provide a feedback signal corresponding to an angular position of the motor **70**. According to the illustrated embodiment, a control cabinet **41** is provided in the machine room. The control cabinet **41** may include a motor drive **40** to control operation of the motor and a separate controller **73** providing instructions to the motor drive **40**. A junction box **74** may be mounted to the top of a housing **72** of the motor **70**, and electrical conductors **76** may run between the control cabinet **41** and the junction box **74**, the motor **70**, the brake **60**, and the encoder **80** to connect the motor drive **40** and the controller **73** with the motor, brake, and encoder. The electrical conductors **76** may conduct electrical power and control signals to or feedback signals from the motor **70**, the brake **60**, and the encoder **80** as will be further described.

(17) Referring also to FIG. 2, the motor drive **40** includes a power conversion section **43** and a control section **45**. The power conversion section **43** converts the input power **21** to the desired voltage at the output **22**. According to the illustrated embodiment, the power conversion section **43** includes a rectifier section **42** and an inverter section **46**, converting a fixed AC input **21** to a variable amplitude and variable frequency AC output **22**. Optionally, other configurations of the power conversion section **43** may be included according to the application requirements. The rectifier section **42** is electrically connected to the power input **21**. The rectifier section **42** may be either passive, such as a diode bridge, or active, including controlled power electronic devices such as transistors. The rectifier section **42** converts the AC voltage input **21** to a DC voltage present on a DC bus **44**. The DC bus **44** may include a bus capacitance **48** connected across the DC bus **44** to smooth the level of the DC voltage present on the DC bus **44**. As is known in the art, the bus capacitance **48** may include a single or multiple capacitors arranged in serial, parallel, or a combination thereof according to the power ratings of the motor drive **40**. An inverter section **46** converts the DC voltage on the DC bus **44** to the desired voltage at the output **22** for the motor **70** according to switching signals **62**.

(18) The control section **45** receives a command signal, feedback signals and generates the switching signals **62** responsive to the command and feedback signals to achieve desired operation of the motor **70**. The control section **45** includes a processor **50** connected to a memory device **52**. It is contemplated that the processor **50** may be a single processor or multiple processors operating in tandem. It is further contemplated that the processor **50** may be implemented in part or in whole on a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a logic circuit, or a combination thereof. The memory device **52** may be a single device or multiple electronic devices, including static memory, dynamic memory, transitory memory, non-transitory memory, or a combination thereof. The memory device **52** preferably stores parameters of the motor drive **40** and one or more programs, which include instructions executable on the processor **50**. A parameter table may include an identifier and a value for each of the parameters. The

parameters may, for example, configure operation of the motor drive **40** or store data for later use by the motor drive **40**.

(19) A motor control module may be stored in the memory **52** for execution by the processor **50** to control operation of the motor **70**. The processor **50** receives feedback signals, **55** and **57**, from sensors, **54** and **56** respectively. The sensors, **54** and **56**, may include one or more sensors generating signals, **55** and **57**, corresponding to the amplitude of voltage and/or current present at the DC bus **44** or at the output **22** of the motor drive **40** respectively. The processor **50** also receives a position feedback signal **95** from the position sensor **80**, such as an encoder or resolver, mounted to the motor **70**. The switching signals **62** may be determined by an application specific integrated circuit **61** receiving reference signals from a processor **50** or, optionally, directly by the processor **50** executing the stored instructions. The switching signals **62** are generated, for example, as a function of the feedback signals, **55**, **57**, and **95**, received at the processor **50**.

(20) The controller **73** in the control cabinet **41** may similarly include a processor and a memory device. It is contemplated that the processor for the controller **73** may be a single processor or multiple processors operating in tandem. It is further contemplated that the processor for the controller **73** may be implemented in part or in whole on a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a logic circuit, or a combination thereof. The memory device for the controller **73** may be a single or multiple electronic devices, including static memory, dynamic memory, transitory memory, non-transitory memory, or a combination thereof. The memory device for the controller preferably stores parameters for operation of the elevator **10** and one or more programs, which include instructions executable on the processor for the controller **73**.

(21) It is contemplated that a portion of the method for controlling motion of the counterbalancing system as described herein may be implemented in whole or in part in the processor **50** of the motor drive **40**, the controller **73**, or a combination thereof. For ease of discussion, the method will be discussed as being implemented on the processor **50** of the motor drive **40**.

(22) Referring next to FIG. **3**, a partial isometric view of an exemplary encoder which may be used in the system of FIG. **1** is provided in accordance with an embodiment of the invention. An encoder **80** may be coupled to a rotor or to a drive shaft **111**, which is, in turn coupled to the rotor. A mounting bracket **84** may be provided to secure a housing **82** of the encoder **80** to a mounting surface **113** provided in the system, such as a mounting surface **113** proximal to the drive sheave **78**. The encoder **80** may be, for example, a high-resolution shaft mounted encoder, such as an EnDat® interface bidirectional rotary encoder available from Heidenhain GmbH of Germany. The encoder **80** includes a transducer in communication with the drive shaft **111** to convert the rotary motion of the drive shaft **111** into an electric signal **95**. The electric signal **95** may be a series of pulses, a sinusoidal waveform, or a serial data word containing the angular position information. The encoder **80** also includes an electronic circuit configured to transmit the electric signal **95** from the transducer to the motor drive **40** via an encoder cable. As a result, the encoder **80** is operative to detect motion of the rotor or drive shaft **111** by detecting angular movement. The illustrated encoder **80** is shown by way of example and is not intended to be limiting. Optionally, a resolver or another position feedback device, providing a signal corresponding to an angular position of the rotor or drive shaft **111**, may be utilized in place of the encoder **80**. In other embodiments, additional and/or alternative encoders, resolvers, or position sensors may be used, and in the same or alternative locations, within the scope of the invention.

(23) Referring now to FIG. **4**, a partial cutaway view of a first exemplary brake **60A** which may be used in the system of FIG. **1** is provided in accordance with an embodiment of the invention. The illustrated brake **60A** includes a toothed hub **102** and a grooved brake rotor **104**. The toothed hub **102** may have a central aperture **106** for mounting to a drive shaft **110** extending from the motor rotor. A notch **108** in the hub **102** is configured to receive a motor key **109**, where the motor key **109** is also fit in a keyway **107** on the motor shaft **110**. The hub **102** is mounted to the motor shaft

110 and the motor key **109** engages both the notch **108** and the keyway **107** to prevent slippage of the hub **102** on the motor shaft **110**. The grooved brake rotor **104** is then slid on to the toothed hub **102** as the brake **60** is mounted to the motor **70**. The brake rotor **104** rotates with the toothed hub during operation of the motor **70**.

(24) The brake **60A** is operative in one of two states. During a first, unenergized state, the brake is set and prevents rotation of the motor **70**. In the first state, one or more springs **112** press an armature disk **114** against a friction plate **105** mounted to the grooved rotor **104**, thereby holding the grooved rotor **104** stationary. As a result, the friction plate **105** holds the grooved rotor **104** and, in turn, the hub **102** stationary. Thus, when the brake is unenergized, the springs **112** cause the brake **60** to be set and hold the drive shaft **110** stationary.

(25) During a second, energized state, the brake **60A** is released and allows rotation of the motor **70**. In the second state, power is provided to an electromagnetic coil **117** wound on a core **118** within the brake **60** in order to establish a magnetic field. The magnetic field is of sufficient strength to attract the armature disk **114** to the core **118** against the force resulting from compressing the spring **112**. Drawing the armature disk **114** toward the core disengages the disk from the friction plate **105** on the grooved rotor **104**. With the brake **60** in the released state, the grooved rotor **104** is able to rotate, which, in turn, allows the hub **102** and drive shaft **110** to rotate.

(26) Referring next to FIG. 5, a perspective view of a second exemplary brake **60B** which may be used in the system of FIG. 1 is provided in accordance with an embodiment of the invention. The brake **60B** includes a pair of arms **140** where one arm **140** is mounted on either side of the drive sheave **78**. A portion of the outer periphery of the drive sheave **78** includes a smooth surface against which brake pads **150** engage rather than grooves configured to receive the ropes **20** of the elevator. Optionally, a brake drum may be mounted adjacent to or on an opposite side of the motor **70** from the drive sheave **78**, where the brake drum provides the smooth surface against which brake pads **150** engage. Each arm **140** has a first end **142** and a second end **144**. The first end **142** of each arm **140** includes a pivot mount **146** and the second end **144** of each arm is mounted to a rod **165** or cylinder. The rod **165** or cylinder is movable between a first position and a second position by an actuator **160**.

(27) The brake **60B** is operative in one of two states. During a first, unenergized state, the brake is set and prevents rotation of the motor **70**. In the first state, one or more springs **155** mounted within or engaging a portion of each arm **140** proximate the second end **144** of each arm apply a force causing each arm **140** to pivot around the pivot mount **146** toward the sheave **78** or brake drum. The force of the springs **155** is sufficient to cause each brake pad **150** to engage the smooth surface of either the drive sheave **78** or the brake drum, preventing rotation of the elevator drive. The springs **155** continually apply the force, such that when the brake is unenergized, the springs **155** cause the brake **60** to be set and hold the drive sheave **78** stationary.

(28) During a second, energized state, the brake **60B** is released and allows rotation of the motor **70**. In the second state, power is provided to the actuator **160**. In the illustrated embodiment, the actuator **160** is a solenoid which causes the rod **165** to extend a distance from within the housing of the solenoid **160**. The distance the rod **165** extends is sufficient to pivot each arm **140** away from the drive sheave **78** or brake drum, allowing rotation of the sheave **78** or brake drum. The rod **165** is threaded as it extends through the second end **144** of one of the arms **140**. Nuts **170** mounted along the threaded portion of the rod **165** may be used to adjust the distance the second end **144** of the arm moves when the solenoid **160** is energized or de-energized. The nuts **170** adjust the amount of force applied by the brake pads **150** on the drive sheave **78** or brake drum when the solenoid **160** is de-energized and adjust the distance the arm **140** rotates, releasing the pressure from the brake pads **150** on the drive sheave **78** or brake drum when the solenoid **160** is energized. Thus, with the brake **60B** properly adjusted, the drive sheave **78** should freely rotate, allowing the motor **70** to control operation of the elevator cab **14**.

(29) The illustrated brakes are intended to be exemplary only and are not limiting. Other

configurations of the brake may include, for example, disk-style brakes engaging a surface of the sheave. The invention discussed below may be used with brakes of various configurations to detect whether the brake has fully opened.

(30) In operation, the motor drive **40** controls operation of the motor **70** which, in turn, controls the elevator drive train and motion of the elevator cab **14**. For discussion herein, it is contemplated that an elevator controller **73** issues a run command to the motor drive **40**. The motor drive **40**, in turn, generates an output signal to the brake **60** to energize or de-energize, synchronizing control of the brake **60** with operation of the motor **70**. According to another embodiment, the elevator controller **73** may generate the output signal to the brake **60** and synchronize control of the brake **60** with operation of the motor drive **40**. In either embodiment, the motor drive **40** may be configured to detect a dragging brake during operation of the motor **70**.

(31) Turning next to FIG. **6**, a flow diagram **200** illustrates one embodiment of the steps performed by a motor drive **40** to detect the dragging brake. At step **202**, the motor drive **40** receives a command to begin a run. During a run, the elevator cab **14** is commanded to move from a first floor, at which it presently located, to a second floor, where passengers either wish to exit or enter the cab **14**. The motor drive **40** is enabled, as shown in step **204**, and begins generating a current corresponding to a desired torque required to hold the elevator cab **14** at its present position.

(32) The amount of torque required to hold the elevator cab **14** at its initial position is a factor of a number of variables. These include, for example, the mass of the car, the mass of the passengers present in the car, the mass of the counter weight, the radius of the drive sheave **78**, and a gear reduction ratio when a gearbox is utilized. Numerous techniques are used in the art to determine the amount of torque required to initially hold the weight of the cabin and passengers present in the cabin. Any suitable technique to determine the required torque may be utilized without deviating from the scope of the present invention. For purposes of discussion, the elevator system may include a weight measurement device that determines a total weight of the cabin **14** and passengers before commanding the motor drive **40** to begin operation. The elevator controller **73** may receive a feedback signal from the weight measurement device and transmit the weight directly or, alternately, may perform some initial processing of the measured weight and transmit a desired torque to the motor drive **40**.

(33) Once the motor drive **40** is enabled and generating a suitable torque to hold the elevator cab **14** at its current position, the motor drive sets the output signal to release the brake, as shown in step **206**. With reference to the brake **60B** illustrated in FIG. **5**, the output signal may be a digital output signal which, for example, closes a relay which, in turn, connects the solenoid **160** of a brake **60B** to power, energizing the solenoid. There is some time required for the solenoid to energize, the rod **165** to extend, and the brake pads **150** to fully disengage the drive sheave **78** or brake drum, thereby disengaging the mechanical holding brake **60B**. As shown in step **208** of FIG. **6**, the motor drive **40** will execute a loop until the brake **60B** opens. The brake **60B** may include a proximity sensor or a microswitch that is set when the arm **140** is fully pivoted away and which generates a brake open feedback signal. Optionally, the motor drive **40** may execute a delay timer corresponding to an expected time required for the mechanical brake **60** to open.

(34) Throughout the duration of time required to open the brake **60**, the motor drive **40** may be executing a motor control module in the processor **50**. The motor drive **40** receives the position feedback signal **95** from the position sensor **80** and holds the elevator cab at its starting position. If the current is too low or too great to hold the elevator cab at its starting position, the motor control module executing on the processor **50** increases or decreases the amount of current output from the motor drive **40** to the motor **70** such that the motor is able to maintain the elevator cab **14** at the desired position. The motor control module may include a single control loop or multiple control loops executing in parallel or in cascade. According to one embodiment of the invention, it is contemplated that a first control loop, operating at a slower update rate, operates on the position feedback signal **95**. The first control loop receives a position reference signal, which corresponds to

the initial position of the elevator cab **14**, and the position feedback signal **95** and determines a difference between the two values. A position control loop including a proportional controller, an integral controller, a differential controller, or a combination thereof, outputs a torque reference signal, corresponding to a desired torque to be applied by the motor **60** to hold the elevator cab **14** at its initial position. The torque reference signal is, in turn, provided as an input to a current control loop operating at a faster update rate. Optionally, the torque reference signal may first be converted to a current reference signal by a torque constant. According to still another option, the torque constant may be incorporated into the controller gains of the current control loop. The current control loop includes a proportional controller, an integral controller, a differential controller, or a combination thereof and regulates the current supplied from the motor drive **40** to the motor **70** to obtain the desired torque at the motor.

(35) Once the brake **60** has opened, the motor drive **40** determines a first value of torque being produced by the motor **70**, as indicated at step **210**. The first value of torque may correspond to the torque reference being generated by the motor control module as indicated above. Optionally, a current feedback signal **57** from a current sensor **56** at the output **22** of the motor drive **40** may be used to determine the amount of torque being produced by the motor **70**. This first value of torque corresponds to the amount of torque required to hold the elevator cab **14** at its current position or, in other words, the amount of torque required to prevent acceleration due to gravity from causing the elevator cab to move either upward or downward in the elevator shaft.

(36) After determining this first value of torque, the motor drive **40** may begin accelerating the motor **70** and, in turn, the elevator cab **14** to a desired speed, as shown in step **212**. The motor drive **40** will continue looping at step **214** until the motor **70** and cab **14** reach a constant speed. Once the motor **70** has reached a constant speed of operation, the motor drive **40** determines a second value of torque being produced by the motor, as shown in step **216**. Just as with the first value of torque, the second value of torque may correspond to the torque reference being generated by the motor control module as indicated above. Optionally, a current feedback signal **57** from a current sensor **56** at the output **22** of the motor drive **40** may be used to determine the amount of torque being produced by the motor **70**. This second value of torque corresponds to the amount of torque required to maintain the constant speed of operation of the elevator cab **14**. In an ideal system, the only external force acting on the elevator cab **14** would be gravity. As a result, the amount of torque required by the motor **70** to maintain the constant speed of operation would be identical to the amount of torque required by the motor **70** to maintain zero speed operation.

(37) In a typical elevator application, some additional torque is required, for example, to overcome friction forces between the cables **20** and sheave **78** or to overcome gearbox inefficiencies. The motor drive **40** may include a parameter stored in memory **52** defining an expected amount of additional torque required by the motor **70** during constant speed operation versus zero speed operation. This value may be set by a technician during commissioning and may define an acceptable difference between the first torque value and the second torque value. Optionally, the motor drive **40** may be configured to automatically detect the difference between the first value of torque and the second value of torque. The motor drive **40** may be programmed to enter a learning mode. During the learning mode, the motor drive **40** may determine one or more values of the difference between the first value of torque and the second value of torque and identify an expected offset between the two values. The motor drive may store this offset and use it in subsequent runs to detect a dragging brake.

(38) As shown in steps **218** and **220**, the motor drive **40** uses the first and second values of torque, previously determined as indicated above, to detect whether the brake **60** is dragging. The motor drive **40** compares the first value of torque to the second value of torque. The comparison may be made, for example, by finding a difference between the first and second values of torque. The difference may, in turn, be compared to a predefined threshold or to the offset value. If the difference between the two values is less than the threshold or less than the offset value, the motor

drive **40** determines that the brake **60** is fully open and moves to step **222**, completing the check for a dragging brake. If, however, the motor drive **40** determines that the difference between the two values is greater than the threshold or the offset value, the motor drive determines that the brake **60** is dragging, as indicated in step **220**. The motor drive **40** may set a fault message alerting a technician to the dragging brake so that repair may be performed before there is excessive heating and/or wear on the brake **60**.

(39) With reference next to FIGS. 7A and 7B, exemplary waveforms illustrating both normal operation and a dragging brake condition are shown. FIG. 7A first shows an exemplary elevator run in both directions. During the first time period shown, the motor drive **40** executes steps **202** through **208** shown in FIG. 6. The motor drive **40** accelerates to a constant speed during the second time period and runs at a constant speed during the third time period. After running for some time at constant speed, the motor drive **40** decelerates back to zero speed and the brake is set until the next run. The steps are then repeated for the opposite direction.

(40) As may be observed in FIG. 7B, the torque required to hold the motor **70** at zero speed during the first time period and to run at constant speed during the third time period are identical when the brake is open, as shown in plot **235**. Although illustrated as requiring no torque, this is a special condition in an elevator at which the weight of the cabin **14** and passengers equal the weight of the counter weight. During this operating condition, no or very little torque is required from the motor **70** to maintain zero speed. If the cabin **14** is empty (minimum weight) or full (maximum weight), the torque required to maintain zero speed will shift up or down around zero. However, the plot **235** would shift by an equal amount for the entire duration of the plot. The balanced condition shown in FIG. 7B is used for ease of illustration and is not intended to be limiting. If the brake is dragging, as shown in plot **240**, the torque required during the first time period remains at zero, but additional torque is required during the third time period to overcome the dragging brake. The motor drive **40** detects the additional torque and provides an indication that the brake is dragging.

(41) In addition to detecting a dragging brake on an individual run, it is contemplated that the present invention may be utilized to detect a change in operation of the brake over time. As previously indicated, a technician may enter an expected difference between torque required by the motor to operate at zero speed versus operating at constant speed. Alternately, the motor drive may execute a commissioning run or monitor one or more runs during initial operation. The motor drive **40** may store an initial value in memory **52** corresponding to an expected difference in torque during operation at constant speed versus operation at zero speed. This initial value may be compared to subsequent runs to detect changes in the torque over time. While the motor drive **40** may still maintain a first threshold indicating that the brake is dragging, the motor drive **40** may also include a second threshold indicating maintenance is required. This second threshold may be set at a lower difference than the first threshold. A separate message may be generated to alert a technician that maintenance is required. The maintenance may then be scheduled at a convenient time.

(42) It should be understood that the invention is not limited in its application to the details of construction and arrangements of the components set forth herein. The invention is capable of other embodiments and of being practiced or carried out in various ways. Variations and modifications of the foregoing are within the scope of the present invention. It also being understood that the invention disclosed and defined herein extends to all alternative combinations of two or more of the individual features mentioned or evident from the text and/or drawings. All of these different combinations constitute various alternative aspects of the present invention. The embodiments described herein explain the best modes known for practicing the invention and will enable others skilled in the art to utilize the invention.

(43) In the preceding specification, various embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the

broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Claims

1. A system for detecting a dragging brake in an elevator drive, the system comprising: an elevator brake, including: at least one braking surface configured to engage a rotating member of the elevator drive to prevent rotation of the elevator drive, a spring configured to apply a force to the at least one braking surface causing the at least one braking surface to engage the rotating member of the elevator drive, and an actuator selectively activated to apply a counter force to the spring, wherein when the counter force is applied to the spring, the at least one braking surface disengages the rotating member of the elevator drive, allowing rotation of the elevator drive; a motor operably connected to cause rotation of the elevator drive; and a motor drive configured to control operation of the motor, wherein the motor drive further comprises: a memory configured to store instructions, and a processor configured to execute the instructions stored on the memory to: determine a first value of torque when the actuator of the elevator brake initially disengages the at least one braking surface from the rotating member of the elevator drive and prior to causing rotation of the elevator drive, determine a second value of torque when the motor is rotating at a constant speed, and compare the first value of torque to the second value of torque to detect when the at least one braking surface of the elevator brake does not fully disengage the rotating member of the elevator drive during rotation of the elevator drive.
2. The system of claim 1 wherein the motor drive is configured to generate a torque reference value corresponding to a desired level of torque supplied by the motor, and wherein the first and second values of torque are torque reference values.
3. The system of claim 1 wherein the motor drive further comprises at least one current sensor configured to generate a current feedback signal corresponding to current output from the motor drive to the motor and wherein the first and second values of torque are determined as a function of the current feedback signal.
4. The system of claim 1, wherein: comparing the first value of torque to the second value of torque includes determining a difference between the first and second values of torque, and detecting when the at least one braking surface of the elevator brake does not fully disengage the rotating member of the elevator drive during rotation of the elevator drive includes comparing the difference to a predetermined threshold and detecting the dragging brake when the difference is greater than the predetermined threshold.
5. The system of claim 1, wherein the motor drive is further configured to initially determine an offset value between the first value of torque and the second value of torque.
6. The system of claim 5, wherein: comparing the first value of torque to the second value of torque includes determining a difference between the first and second values of torque, and detecting when the at least one braking surface of the elevator brake does not fully disengage the rotating member of the elevator drive during rotation of the elevator drive includes comparing the difference to the offset value and detecting the dragging brake when the difference is greater than the offset value.
7. A system for detecting a dragging brake for an electric motor, the system comprising: a motor configured to control motion of a drive train for a load having an additional force acting on the load to cause motion of the load when not controlled by the drive train; a motor drive operatively connected to the motor, wherein the motor drive is configured to: determine a first value of torque when a holding brake in the drive train is initially released and prior to causing motion of the drive train with the motor, determine a second value of torque when the motor is rotating at a constant speed, and compare the first value of torque to the second value of torque to detect when the holding brake does not fully disengage the drive train during operation of the motor.

8. The system of claim 7 wherein the motor drive is configured to generate a torque reference value corresponding to a desired level of torque supplied by the motor, and wherein the first and second values of torque are torque reference values.
9. The system of claim 7 wherein the motor drive further comprises at least one current sensor configured to generate a current feedback signal corresponding to current output from the motor drive to the motor and wherein the first and second values of torque are determined as a function of the current feedback signal.
10. The system of claim 7, wherein: comparing the first value of torque to the second value of torque includes determining a difference between the first and second values of torque, and detecting when the holding brake does not fully disengage the drive train includes comparing the difference to a predetermined threshold and detecting the dragging brake when the difference is greater than the predetermined threshold.
11. The system of claim 7, wherein the motor drive is further configured to initially determine an offset value between the first value of torque and the second value of torque.
12. The system of claim 11, wherein: comparing the first value of torque to the second value of torque includes determining a difference between the first and second values of torque, and detecting when the holding brake does not fully disengage the drive train includes comparing the difference to the offset value and detecting the dragging brake when the difference is greater than the offset value.
13. The system of claim 7, wherein: the motor drive further comprises an input configured to receive a position feedback signal, the position feedback signal is generated by a position feedback device operatively connected to the motor, the position feedback signal corresponds to an angular position of the motor, and the motor drive is configured to control operation of the motor to maintain the load at an initial position determined from the position feedback signal when the holding brake is released and until the first value of torque is determined.
14. The system of claim 7, wherein: the drive train is an elevator drive train, the holding brake is an elevator brake, including: at least one braking surface configured to engage a rotating member of the elevator drive train to prevent rotation of the elevator drive train, a spring configured to apply a force to the at least one braking surface causing the at least one braking surface to engage the rotating member of the elevator drive train, and an actuator selectively activated to apply a counter force to the spring, wherein when the counter force is applied to the spring, the at least one braking surface disengages the rotating member of the elevator drive train, allowing rotation of the elevator drive train, the load is an elevator cab and counter weight, and the additional force is a force resulting as a difference between a weight of the elevator cab and a weight of the counter weight.
15. A method for detecting a dragging brake for an electric motor, the method comprising the steps of: determining a first value of torque generated in a motor operatively connected to a motor drive when a holding brake in a drive train between the motor and a load controlled by the motor is initially released, wherein: the motor drive is configured to determine the first value of torque, and the load has an additional force acting on the load to cause motion of the load when not controlled by the drive train; determining a second value of torque when the motor is rotating at a constant speed; and comparing the first value of torque to the second value of torque to detect the dragging brake.
16. The method of claim 15 further comprising the step of generating a torque reference value with the motor drive, wherein the torque reference value corresponds to a desired level of torque supplied by the motor and wherein the first and second values of torque are torque reference values.
17. The method of claim 15 further comprising the step of generating a current feedback signal with at least one current sensor in the motor drive, wherein the current feedback signal corresponds to current output from the motor drive to the motor and wherein the first and second values of torque are determined as a function of the current feedback signal.

18. The method of claim 15 wherein the step of comparing the first value of torque to the second value of torque further comprises the steps of: determining a difference between the first and second values of torque, and comparing the difference to a predetermined threshold, and detecting the dragging brake when the difference is greater than the predetermined threshold.

19. The method of claim 15, further comprising an initial step of determining an offset value between the first value of torque and the second value of torque with the motor drive, wherein the step of comparing the first value of torque to the second value of torque further comprises the steps of: determining a difference between the first and second values of torque, and detecting the dragging brake when the difference is greater than the offset value.

20. The method of claim 15, further comprising the steps of: receiving a position feedback signal at an input of the motor drive, wherein the position feedback signal is generated by a position feedback device operatively connected to the motor and wherein the position feedback signal corresponds to an angular position of the motor, and controlling operation of the motor with the motor drive to maintain the load at an initial position determined from the position feedback signal when the holding brake is released and until the first value of torque is determined.
